

Introduction to Futures

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Lecture review

Definition of Futures

Futures contract is a **standardized** agreement that fixes terms of an exchange that will take place at some future date.

Terms of an exchange:

- ① underlying asset
- ② contract size
- ③ delivery month
- ④ price quote
- ⑤ delivery arrangements

Underlying Asset

Futures price derived from its underlying asset's price → main feature of a **DERIVATIVE**

Main types of underlying assets:

- Indices CME volumes
- Commodities
- Equities
- Currencies
- Interest rates

Commodity quality might differ ⇒ grade of quality in contract specification

Contract Size

Contract unit or Multiplier

- 1 000 barrels of crude oil
- 100 pieces of PX index
- 25 000 lb. of copper
- ...

Price quotation or minimum **tick**

- a minimum amount that the price of an underlying asset can fluctuate upward or downward and will influence profit on future contract
- set by the exchange and varies by contract
- e.g. S&P 500 – 1/4 of index point, 1 index point e.g. 50 USD $\Rightarrow$ 1 tick move is 12.5 USD
- WTI Crude oil – 1 cent, contract for 1000 barrels $\Rightarrow$ 1 tick move is 10 USD

Minimum gain/loss:

- $\Rightarrow \text{Tick} * \text{Multiplier}$

Delivery Month

⇒ Usual maturities range from one month to two years

⇒ Financial contracts (interest rates, forex and stock indexes) expire quarterly, in *March, June, September and December*

⇒ Commodity contract are available for *each month*

Price Quote

Tick size differs according to the underlying asset

Example:

- *crude oil is quoted in \$/barrel to two decimal places,*
- *treasury bonds are quoted in dollars and thirty-seconds of dollar (in $\frac{1}{32}$) $\Rightarrow$ US T-Bill quote price:*
 - *$99-21\frac{3}{4}$ translates to $99 + \frac{21.75}{32} = 99.68$*

Price quote depends on:

- valuation rules of the underlying asset
- contract size

Delivery Arrangements

⇒ Only in case of physical delivery

⇒ The place where the delivery takes place must be specified; particularly important for commodities that involve transport costs

⇒ In the case of alternative delivery locations, price might be adjusted

Settlement of a Future Contract

- ① Terminating the contract before the delivery date
 - Long position (future buyer) offset by a short position (selling the future) and vice versa
- ② Holding the position until the delivery date
 - Physical settlement: exchange of the underlying asset and cash
 - Cash settlement: financial compensation

Less than 1% of all futures contracts end with delivery of the commodity!

⇒ Nearly all traders reverse their trades (called "offset") before the contract expiration date.

Spot vs. Future Price of an Asset

Can differ before the delivery date according to

- Market expectations
- Cost of carry - cost of holding a position
 - e.g. opportunity costs
- Expected payments before the delivery date

Base = Spot price - Future price

- *Contango*: **negative** Base
- *Backwardation*: **positive** Base

⇒ Equal at delivery date

Margining

Purpose: **minimize credit risk**

Objective for both future buyer and seller

Expressed in % of contract value (usually 5-15%)

Clearing house role in margining:

- each party of a futures contract has to have a margin account by a clearing house (non-members have to channel their trades through members)
- calculates net positions of members

Marking-to-market

- daily settlement price is established for every futures contract and set by the exchange
- final daily settlement price $\Rightarrow$ broker calculates profit and loss from the previous day
- no losses are carried forward $\Rightarrow$ must be settled every day
- changes influence the margin account balance
- financial safeguard

Types of margin

Futures margin = amount of money you have to deposit and keep at your broker

① *Initial margin:*

- percentage of the purchase price of securities that the investor must pay by cash or marginable securities
- deposited to investor's margin account
- amount required to **initiate** the position

② *Variation margin:*

- calculated and collected on a daily or intraday basis to reduce the exposure of carrying highly risky positions
- if the value of the contract has increased, the trader with long position received funds (variation margin) from the trader with the short position and vice versa
- $\text{Variation Margin} = \text{Initial Margin} - \text{Margin balance}$

③ *Maintenance margin:*

- minimum amount that must be **maintained** on a margin account
- payment that has to be deposited on the margin account in case of adverse price movements of the futures contracts that these members

Price Limits and Position Limits

Daily price limits → specified by the exchange

⇒ **Purpose:**

- prevent **large** price movements because of speculation excess
 - ⇒ situation usually results into a cease of trading

Position limit → maximum number of contracts that a speculator may hold

⇒ **Purpose:**

- prevent undue influence on the market

link1, link2

- Sell here 30 april at 142
- 1 contract = 15000 pounds
- offer: 450 000 pounds at 1.42 = 639 000 dollars
- closed trade at 29 cents = 1.13 per pound profit
- 508 500 dollars profit on that trade
- since 2010 The Eddie Murphy Rule

Exercises

Problem 1

- ① In January 9, John Doe purchased ten June futures contract on the Nasdaq 100 Index. He decides to close out his position in April 16.
- ② Jane Doe is a futures trader. In early August, she took a short position in an S&P 500 Index futures contract expiring in September. After a week, she decides to close out her position.

⇒ **Describe in details, how they would do so.**

Problem 2

A company enters into a short futures contract to sell 5 000 bushels of wheat for 250 cents per bushel. The initial margin is \$3 000 and the maintenance margin is \$2 000.

- What price change would lead to a margin call?
- Under what circumstances could \$1 500 be withdrawn from the margin account?

Problem 3

A copper futures contract requires the long trader to buy 25000 lbs of copper. A trader buys one November copper futures contract at a price of \$0.78/lb.

→ *Theoretically, what is the maximum loss this trader could experienced?*

Another trader sells one November copper futures contract.

→ *Theoretically, what is the maximum loss this trader with a short position could have?*

Problem 4

Fill in the following table:

Day (1)	Margin Account Beginning Balance (2)	Funds Deposited (3)	Settlement Price (4)	Futures Price Change (5)	Gain/Loss (6)	Margin Account Ending Balance (7)
0			82.00			
1			84.00			
2			78.00			
3			81.00			
4			79.00			

- Initial futures price = \$82/contract
- Initial margin requirement = \$5/contract
- Maintenance margin requirement = \$2/contract
- # of long contracts = 20

Problem 5

At the end of one day a clearinghouse member is long 85 contracts, and

- the settlement price is \$42 500/contract.
- the original initial margin is \$2 000/contract.

On the following day the member becomes responsible for clearing an additional 15 long contracts, entered into at a price of \$43 750/contract. The settlement price at the end of this day is \$43 000.

How much does the member have to add to its margin account (in total including the initial margin) with the exchange clearinghouse?

Problem 6

An investor enters into two long futures contracts on frozen orange juice. Each contract is for the delivery of 15,000 pounds. The current futures price is 160 cents per pound, the initial margin is \$6,000 per contract, and the maintenance margin is \$4,500 per contract.

- What price change would lead to a margin call?
- Under what circumstances could \$2,000 be withdrawn from the margin account?